

Vishal S

Master of Technology (Software Engineering)
Vellore Institute of Technology, Chennai

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EDUCATION

Integrated M.Tech (Software Engineering)

Vellore Institute of Technology, Chennai

2022 – 2027

Nazareth Matriculation Higher Secondary School, Chennai

Class XII: 92.50%

CGPA: 8.57 / 10

2022

Class X: 88.20%

CERTIFICATIONS

Oracle Cloud Infrastructure (OCI) AI Foundations Associate

Oracle University

- Covered Machine Learning, Generative AI, Large Language Models (LLMs), Multimodal Generative AI, and AI-driven analytics workflows
- Gained exposure to cloud-based data processing, scalable AI deployment, and model lifecycle management

Agile Scrum Practices

Infosys Springboard

- Learned Agile methodologies, sprint planning, iterative development, collaborative debugging, and software delivery practices

PROJECTS

LLM-Powered AI Voice Agent System with RAG Pipelines

Python, FastAPI, LangChain, FAISS

- Developed modular backend APIs and asynchronous pipelines for real-time AI query processing using Python and FastAPI
- Designed and tested REST API endpoints using Postman for reliable request/response validation
- Built maintainable API-driven components following Agile development and collaborative Git workflows
- Implemented context-aware RAG retrieval using FAISS vector similarity search for efficient conversational memory handling
- Debugged backend API workflows, identified performance bottlenecks, and improved response stability, reducing average query response time by approximately 30%

Medical AI Assistant — Healthcare Inventory & Procurement SaaS Platform

GoLang, ReactJS, SQL, Docker

- Built a full-stack healthcare SaaS platform with a GoLang/Gin backend, Next.js frontend, JWT authentication, and RESTful API workflows for procurement, supplier management, and inventory tracking
- Engineered SQL-based data models, Redis caching pipelines, and Dockerized deployment workflows to support scalable application architecture
- Developed AI-powered inventory insights and low-stock prediction modules with responsive dashboard analytics and reusable ReactJS component architecture

Code Sage — Gamified Programming Learning Platform (Android App)

Java, Android, REST APIs, Postman

- Developed a Java-based Android platform with quiz modules, analytics dashboards, and REST API integrations
- Applied Object-Oriented Programming principles and modular architecture while validating API functionality using Postman
- Debugged API integration and frontend issues to improve application stability and user experience
- Built maintainable application components using Git, unit testing, and Agile sprint-based delivery practices

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

Backend & APIs: FastAPI, GoLang (Gin), REST APIs, API Testing

Frontend: ReactJS, Next.js, HTML, CSS

Databases: MySQL, PostgreSQL, Redis

AI/ML: Generative AI, LLMs, RAG, Machine Learning

Cloud & DevOps: Docker, Git, CI/CD, OCI

Software Engineering: Object-Oriented Programming, Software Testing, Debugging, Agile, Code Reviews

PATENTS & PUBLICATIONS

A System for Capturing Handwritten Content on a Board and a Method Thereof

Published: July 2026

Indian Patent Application No. 202641091219

- Co-invented and published a patent for a novel IoT-based interactive board system enabling real-time capture and digitization of handwritten content
- Engineered the core embedded software in C++, applying OOP principles and efficient algorithms for reliable real-time performance
- Collaborated with faculty and student co-inventors through the complete filing and publication process